

# S<sup>4</sup>D

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**We introduce the first reinforcement-learned routing policy for hierarchical speculative decoding, and set a new state of the art.**

VIA-SD adds a slim intermediate verifier, but routes between tiers using confidence thresholds tuned by hand.

We present **three key contributions**:

1. A **learned per-token gating policy** that recasts tier-routing as a sequential decision problem, so a tiny  $q$ -free MLP replaces the fixed thresholds.
2. A **reward** that balances output quality against the latency of expensive verifier calls.
3. A **per-token RL formulation** with principled credit assignment (REINFORCE, then GRPO, then a per-token advantage).

**The payoff:** we match *lossless* accuracy (**0.91**) at  **$4\times$  fewer** full-model calls,  **$2.2\times$  faster** than greedy. We **Pareto-dominate** the paper's VIA-SD at roughly  **$2\times$  its speedup** and  **$2.3\times$  its accuracy**, and reach up to  **$3.5\times$**  when pushed for speed.

# The bottleneck: LLM decoding is slow and sequential

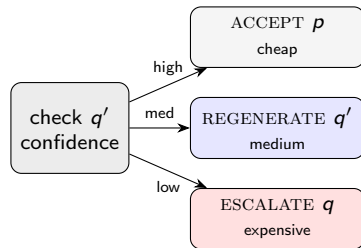
- Generating each token streams the entire model from memory, so LLM inference is memory-bandwidth bound and dominated by sequential forward passes.
- Speculative decoding fixes this: a cheap *drafter* guesses several tokens, and the big *verifier* checks them all in one parallel pass.
- Accepted guesses give many tokens per expensive forward. It is **lossless**, and runs about **1.4×** faster on our models.
- The one lever is the **acceptance rate**, because every rejected token forces a full-model step.

# VIA-SD improves on vanilla SD

Vanilla SD makes a **binary** choice per token: accept the drafter, or reject and pay for the full model  $q$ .

VIA-SD adds a **slim middle verifier**  $q'$  (the target  $q$  with about 45% of its layers skipped), giving **three** choices. Borderline “middle-zone” tokens are handled cheaply by  $q'$ , so the full model is called far less often.

Since  $q'$  is the target verifying its *own* draft through a reduced copy of itself, this is the **Semi-Self** in  $S^4D$ .



## But it routes with thresholds that are picked by hand

VIA-SD makes the three-way choice with **two fixed confidence thresholds** on  $q'$ :

$$\alpha_1 = 0.5$$

$$\alpha_2 = 0.3.$$

- They are **global and static**: one cutoff for every token and every context.
- They are **brittle**, and must be re-tuned for each new model and task.
- They are **sub-optimal**: even after a full sweep, accuracy peaks at 0.50 on our setup.

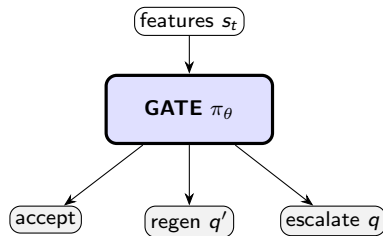
Choosing a tier for each token, from its context, is a sequential decision problem. **So we learn it** — the gate teaches itself. That is the **Self-Taught** in  $S^4D$ .

# Contribution 1/3: a learned per-token gating policy (MDP)

We recast tier-routing as a **Markov decision process**:

- **State**  $s_t$ :  $q$ -free features (drafter/ $q'$  confidence, agreement,  $\text{KL}(p\|q')$ , position).
- **Action**  $a_t \in \{\text{accept}, \text{regenerate}, \text{escalate}\}$ .
- **Policy**  $\pi_\theta(a_t | s_t)$ : a tiny MLP.

The full model  $q$  is consulted only to *score* at train time. It is never a gate input, so the gate is free to run at inference.



## Contribution 2/3: a reward balancing quality and cost

**Two ingredients per token:**

$$\text{quality: } \text{match}_t = \mathbf{1}[\hat{y}_t = \arg \max_v q(v \mid x_{<t})]$$

$$\text{cost: } \text{cost}_t = (t_{p_1} + t_{q'}/\gamma + \mathbf{1}[\text{ESCALATE}] t_q) / t_{q_1}$$

**Reward** (cost in units of one full-model step):

$$\text{trajectory: } R = \overline{\text{match}} - \lambda \overline{\text{cost}} (+ r_c \text{ correct}) \qquad \text{per-token: } r_t = \text{match}_t - \lambda \text{cost}_t$$

$\lambda$  is the quality-versus-speed knob. `match` is a *dense* proxy for the sparse true objective (final-answer correctness).

## Contribution 3/3: per-token RL with credit assignment

**Policy gradient** (all variants, plus an entropy bonus  $\eta \mathcal{H}[\pi_\theta]$ ):

$$\nabla_\theta J = \mathbb{E} \left[ \sum_t A_t \nabla_\theta \log \pi_\theta(a_t | s_t) \right]$$

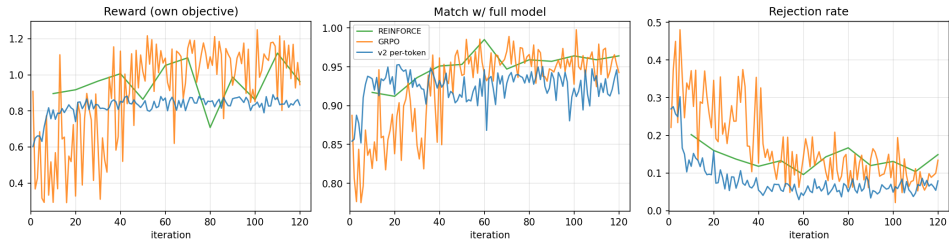
They differ only in the **advantage**  $A_t$ :

- **REINFORCE** (trajectory):  $A = R - b$ , with EMA baseline  $b \leftarrow (1-\tau)b + \tau R$ .
- **GRPO** (group):  $A_i = \frac{R_i - \text{mean}_G R}{\text{std}_G R}$ , with PPO-clip  $\rho_t = \frac{\pi_\theta}{\pi_{\theta_{\text{old}}}}$  and KL to  $\pi_{\text{ref}}$ .
- **per-token (v2)**:  $\hat{A}_t = \frac{r_t - \text{mean}_{\text{batch}} r}{\text{std}_{\text{batch}} r}$ , local credit and the **lowest variance**.



# Training converges, and per-token (v2) is cleanest

RL training curves (0.5B→14B, GSM8K)



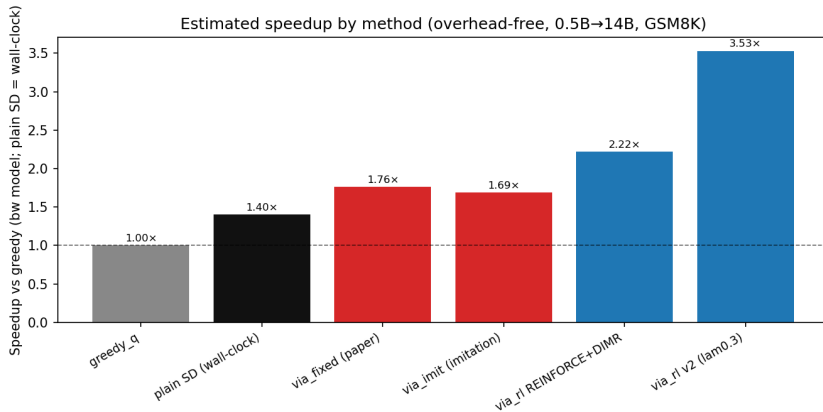
# Results: the learned gate dominates

| Method                        | GSM8K acc    | q-calls/tok  | speedup      |
|-------------------------------|--------------|--------------|--------------|
| greedy- $q$ (reference)       | 0.887        | 1.000        | 1.00×        |
| plain SD (standard, lossless) | 0.907        | 0.259        | 1.40×        |
| via_fixed (paper, tuned)      | 0.40         | 0.31         | 1.76×        |
| via_imit (imitation only)     | 0.34         | 0.29         | 1.69×        |
| via_rl REINFORCE              | 0.880        | 0.243        | 2.29×        |
| via_rl GRPO                   | 0.853        | <b>0.162</b> | 2.94×        |
| via_rl REINFORCE+DIMR         | <b>0.907</b> | 0.250        | 2.22×        |
| via_rl v2 per-token           | 0.713        | 0.104        | <b>3.53×</b> |

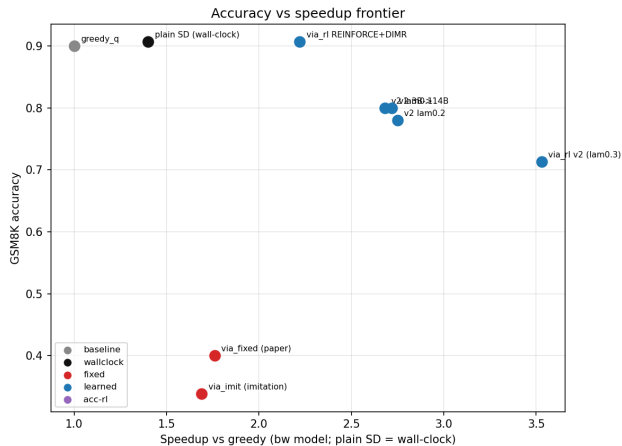
⇒ **S<sup>4</sup>D** is

**state-of-the-art hierarchical speculative decoding.**

# Latency: speedup by method



# The accuracy and speed frontier



# Ablation: which model should we scale?

| Setup                                       | GSM8K acc   | q-calls/tok  | note                                     |
|---------------------------------------------|-------------|--------------|------------------------------------------|
| 0.5B → 14B (main)                           | 0.88–0.91   | 0.10–0.25    | baseline pair                            |
| <b>3B</b> → 14B (bigger <i>drafter</i> )    | 0.80        | <b>0.028</b> | <b>36×</b> <b>fewer</b> full-model calls |
| 0.5B → <b>32B</b> (bigger <i>verifier</i> ) | <b>0.33</b> | 0.07         | accuracy collapses                       |

- A stronger **drafter** raises acceptance, so the gate escalates far less at similar accuracy.
- A bigger **verifier** with the same weak drafter widens the gap, and the accept-heavy gate emits tokens the target would reject.

*The lever is the drafter, not the verifier.*

# Ablation: the slim verifier $q'$ and the cost knob $\lambda$

- **Better  $q'$  via DIMR layer-skip search:** REINFORCE 0.88  $\rightarrow$  0.91. A trustworthy regenerate tier lets the gate use  $q'$  instead of escalating, and the tokens come out right.
- **But  $q'$  quality is moot once the gate is accept-heavy** (the v2 policy routes around the regenerate tier entirely).
- **The  $\lambda$  knob is a real frontier:**  $\lambda=0.1 \rightarrow 0.80$ ,  $0.2 \rightarrow 0.78$ ,  $0.3 \rightarrow 0.71$  accuracy.
- **Honest negative:** optimizing final-answer correctness *directly* (instead of the match proxy) degraded every warm-start we tried; the match-trained REINFORCE+DIMR (0.91) stays best.

# Impact: a new accuracy and efficiency frontier

- **Lossless quality, 4× cheaper.** We match greedy/lossless accuracy (**0.91**) at 4× **fewer** full-model calls, a 2.2× speedup, with *zero accuracy lost*.
- **A tunable frontier.** One knob gives 2.7× at −0.1 accuracy, or 3.5× at −0.19, up to 10× **fewer** full-model calls.
- **Pareto-dominates VIA-SD.** About 2× its speedup *and* +0.3 **to** +0.5 accuracy. Where its fixed thresholds collapse to 0.40, our gate holds **0.71 to 0.91**.
- **Scales with the drafter.** A 3B drafter gives 36× **fewer** full-model calls at 0.80 accuracy, and it is a free-to-deploy tiny MLP.

**S<sup>4</sup>D is the new state of the art in hierarchical speculative decoding.**

*Stop hand-picking thresholds. Let the model schedule its own compute.*

# Future work

- **Spec-spec decoding:** make the drafter itself speculative, a recursive hierarchy where the gate routes across *levels*.
- **KnapSpec:** treat verification as a **knapsack**, spending a fixed budget of expensive calls on the highest-value tokens (value is correctness impact, weight is cost).
- **Correctness-floor RL:** minimize cost *subject to* an accuracy floor, which avoids the reward-hacking collapse.
- **Scale:** bigger drafters, more tasks, and real-engine wall-clock (static cache, CUDA graphs).



# S<sup>4</sup>D

Self-Taught Semi-Self Speculative Decoding

**The new state of the art in hierarchical speculative decoding.**

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Thanks!    questions?